

Ledger Service v3

Migration Readiness Assessment

An assessment of whether the v3.0 ledger can safely replace v2.7 in production, based on twenty-one days of shadow traffic and a full performance soak.

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Sample document. Demonstrates a cover page, an automatic table of contents, running headers and footers with page numbering, multi-level headings, bulleted and numbered lists, styled tables with alternating rows and a totals row, callout boxes, headline metric tiles, a block quotation, footnotes, internal and external hyperlinks, mixed portrait and landscape sections, and explicit page breaks. All names, figures and findings are invented.

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1. Executive summary

Ledger Service v3.0 is materially faster than the version it replaces, and twenty-one days of shadow traffic found no unexplained divergence between the two systems. Posting throughput more than doubled, reversal latency fell by 57%, and the projected infrastructure bill drops by \$3,150 a month even after the worker tier grows.

Three items stand between the current state and a safe cutover. The statement endpoint has regressed and now misses its latency objective by roughly 50%. The rollback path has been designed but never rehearsed against the 30-minute recovery objective. And only one of the two required game days has been run, which leaves half the on-call rotation working from a runbook they have not exercised.

+124% Posting throughput	37 ms p95, POST /entries	37 Mismatches in 412M	-5.7% Monthly run cost
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Recommendation: conditional go. Proceed with the October cutover on the condition that all three blockers listed in §6 are closed and signed off by 25 September. None of them is technically difficult; all of them are schedule risks.

2. Scope and method

This assessment covers the ledger service only. It does not cover the reporting warehouse, the settlement scheduler, or the partner-facing API gateway, all of which have their own migration tracks.

2.1 What was tested

- Functional parity against the v2.7 contract test suite (1,847 cases).
- A 45-minute performance soak per workload, repeated three times per configuration.
- Twenty-one consecutive days of shadow traffic at 100% of production volume.
- A restore drill from the previous night's snapshot into a clean account.
- Two failure injections: primary database failover, and a full cache flush at peak.

2.2 How the numbers were produced

Every figure in this report is the median of three runs against the build named in Appendix A, measured at the load generator rather than inside the service.¹ Percentage changes are computed against the v2.7 baseline captured on the same hardware in the same week, so they are not affected by drift in the underlying instance types.

We agreed early that a number nobody can reproduce is worse than no number at all. Everything here comes with a run identifier and a retained artefact.

— Assessment charter, §2

3. Findings

3.1 Throughput

Write paths improved substantially; the cached read path improved modestly; the cold read path regressed.

Workload	v2.7 baseline	v3.0 candidate	Change
Posting, single entry	1,420 tps	3,180 tps	+124%
Posting, batch of 500	9,600 tps	24,100 tps	+151%
Balance read, cached	41,000 tps	44,500 tps	+8.5%
Balance read, cold	2,050 tps	1,780 tps	-13%
Reversal	880 tps	1,910 tps	+117%

Table 1 — Sustained throughput by workload, median of three 45-minute soaks.

The cold-read regression is a direct consequence of the new cache key layout: entries that used to share a key now occupy separate slots, so a flush costs more to recover from. Pre-warming closes most of the gap, which is why R-06 is mitigated by a staged rollout rather than a code change.

3.2 Latency against objectives

Endpoint	v2.7 p95	v3.0 p95	Target	Verdict
POST /entries	84 ms	37 ms	≤ 50 ms	Pass
GET /accounts/{id}/balance	12 ms	11 ms	≤ 25 ms	Pass
POST /reversals	143 ms	61 ms	≤ 100 ms	Pass
GET /statements (90 d)	1,910 ms	2,240 ms	≤ 1,500 ms	Fail

¹Raw results, including per-run variance, are retained in the performance archive under run identifiers RA-2026-014-01 through -09.

Table 2 — p95 latency at peak load against the published service level objectives.

Blocker B-1. GET /statements over a 90-day range is 17% slower than v2.7 and misses its 1,500 ms objective by 740 ms. The cause is a missing covering index on the new partition scheme, confirmed by query plan diff. The fix is queued for 3.0.5 and must land before Phase 2.

3.3 Data integrity

Over twenty-one days the shadow run compared 412,380,551 posted entries. Thirty-seven mismatches were recorded, all of them on foreign-exchange legs carrying six or more decimal places, and all of them traced to a single rounding path corrected in 3.0.4.² No mismatch was found after the fix shipped on day 14.

- 37 mismatches in 412,380,551 entries — a rate of 9.0×10^{-8} .
- All 37 occurred before 3.0.4; zero in the final seven days.
- Maximum absolute divergence: 0.000004 of the minor unit.
- No mismatch affected a settled balance.

3.4 Operational readiness

Area	Owner	State	Evidence
Runbooks rewritten	S. Okafor	Complete	PE-DOC-311, reviewed 4 Aug
Dashboards + alerts	M. Farouk	Complete	18 panels, 11 alert rules
Rollback rehearsal	A. Reyes	Not started	Blocker B-2
Game days (2 required)	S. Okafor	1 of 2 done	Blocker B-3
Backup / restore drill	J. Lindqvist	Complete	Restored 412M rows in 24 m
Capacity headroom $\geq 40\%$	M. Farouk	Complete	Measured 61% at peak

Table 3 — Readiness checklist as of 21 August 2026.

²The count excludes 1,204 records quarantined by the pre-existing foreign-exchange cutoff job, which behaves identically in both versions.

4. Cost analysis

The API tier shrinks because v3.0 handles more than twice the posting throughput per core. The worker tier grows because reconciliation now runs continuously rather than nightly.

Line item	v2.7 / month	v3.0 / month	Change
Compute — API tier	\$18,400	\$11,900	−\$6,500
Compute — worker tier	\$7,250	\$8,900	+\$1,650
Managed PostgreSQL	\$22,000	\$22,000	\$0
Redis cache	\$3,100	\$4,400	+\$1,300
Object storage and egress	\$1,860	\$1,910	+\$50
Observability	\$2,400	\$2,750	+\$350
Total	\$55,010	\$51,860	−\$3,150

Table 4 — Projected monthly infrastructure cost at current production volume.

One-time migration cost is estimated at 6.5 engineer-weeks, or roughly \$41,000 fully loaded. Against a saving of \$3,150 a month the payback period is thirteen months, which sits inside the eighteen-month threshold the platform group uses for infrastructure work.

Note. These projections hold volume constant. Finance currently forecasts 22% transaction growth over the next year; at that volume the saving widens to about \$4,900 a month because the API tier is the component that scales with posting rate.

5. Risk register

Score is likelihood (1–5) multiplied by impact (1–5). Anything scoring 12 or above requires a named owner and a dated mitigation before cutover.³

ID	Risk	Likelihood	Impact	Score	Owner	Mitigation
R-01	Statement endpoint misses its latency SLO	Likely	Moderate	12	M. Farouk	Add covering index; ship in 3.0.5 before cutover
R-02	Dual-write drift during the cutover window	Possible	Severe	15	A. Reyes	Freeze writes for 90 s, reconcile before opening traffic
R-03	FX rounding defect recurs after 3.0.4	Unlikely	Severe	10	J. Lindqvist	Property-based test in CI across 10 ⁷ generated cases
R-04	Rollback exceeds the 30-minute RTO	Possible	Severe	15	A. Reyes	Rehearse twice; keep v2.7 warm for 14 days
R-05	On-call unfamiliar with the new runbook	Likely	Minor	8	S. Okafor	Two game days completed before cutover
R-06	Cold-read regression under cache flush	Possible	Moderate	9	M. Farouk	Pre-warm the cache; roll out in 5% stages

Table 5 — Open risks at the time of assessment. Two entries score 15 and gate the cutover.

³Scoring follows the platform group's standard risk matrix, PE-STD-004.

6. Recommendations

In priority order. The first three are the blockers referenced throughout this report.

1. **Ship the covering index in 3.0.5** and re-measure GET /statements under the same soak. Do not begin Phase 2 until the endpoint is inside its objective. (Blocker B-1, owner M. Farouk, due 8 September.)
2. **Rehearse the rollback twice** against a production-sized dataset and record the wall-clock time. If either rehearsal exceeds 30 minutes, the cutover date moves. (Blocker B-2, owner A. Reyes, due 18 September.)
3. **Run the second game day** with the half of the rotation that has not yet participated, using the rewritten runbook without assistance. (Blocker B-3, owner S. Okafor, due 25 September.)
4. **Keep the v2.7 fleet warm for fourteen days** after cutover rather than the seven originally planned. The incremental cost is about \$12,800 and it buys a second full billing cycle of confidence.
5. **Add a property-based rounding test** to CI covering ten million generated foreign-exchange cases, so the class of defect found in §3.3 cannot silently return.
6. **Publish the reconciliation dashboard** to the finance team before Phase 2, so that drift is visible to the people most likely to notice a wrong number first.

7. Proposed timeline

Phase	Activity	Window	Exit criterion
0	Prep, code freeze, index deploy	1–11 Sep 2026	3.0.5 tagged and in staging
1	Shadow traffic at 100%	14–25 Sep 2026	Zero unexplained mismatches for 7 days
2	Canary 5% → 25% of live traffic	28 Sep – 9 Oct 2026	No SLO breach over 10 days
3	Full cutover	12–13 Oct 2026	100% traffic on v3.0, drift = 0
4	Decommission v2.7	27 Oct 2026	Fleet terminated, costs reconciled

Table 6 — Phased cutover plan. Each phase has a single, testable exit criterion.

The plan assumes all three blockers close on schedule. If B-2 slips past 18 September the cutover moves to the following maintenance window on 26 October; the team should not compress Phase 2, which exists to catch exactly the kind of slow-burning divergence that a shorter canary would miss. The full operational procedure is in the [cutover runbook](#), and the risks it hedges against are listed in [§5, Risk register](#).

Appendix A — Test environment

All figures in this report were produced against the following environment. It mirrors production topology at one-third scale, with identical instance families.

Component	Specification
Region	eu-central-1, three availability zones
API tier	6 × c7g.2xlarge (8 vCPU, 16 GiB)
Worker tier	4 × c7g.xlarge (4 vCPU, 8 GiB)
Primary database	db.r7g.4xlarge, 2 read replicas, 4 TiB gp3
Cache	3 × cache.r7g.large, cluster mode enabled
Dataset	412,380,551 entries across 8,412,077 accounts
Load generator	k6 v0.52, 45-minute soak, 12 injector nodes
Build under test	ledger-svc 3.0.4+build.2291 (commit 7f3ac1e)

Table 7 — Test environment specification.

Appendix B — Glossary

Term	Definition
Canary	A deployment that receives a small, deliberately capped share of live traffic before a full rollout.
Drift	Any divergence between the records held by the old and new systems while both are being written to.
Dual write	Writing each transaction to both v2.7 and v3.0 so the two can be compared entry by entry.
p95	The value below which 95% of measurements fall; the tail that most users actually notice.
RPO	Recovery point objective — the maximum data loss, expressed as a period of time, that is tolerable.
RTO	Recovery time objective — the maximum tolerable time to restore service after a failure.
Shadow traffic	A copy of live requests replayed against the new system whose responses are discarded.
SLO	Service level objective — an internal target for a measurable aspect of service quality.

Table 8 — Terms used in this report.

End of assessment PE-RA-2026-014 Rev. C. Circulation is limited to the platform group and the office of the VP Engineering.